A Benchmark-Integrity Audit Protocol for Conformal Battery State-of-Health Evaluation: Recent-Label Dependence, Schema-Bundle Sensitivity, and Threshold-Decision Risk

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Abstract

Conformal intervals for lithium-ion battery state-of-health (SOH) inform maintenance decisions, but cross-dataset validation can conceal measurement conditions. We develop a six-gate benchmark-integrity audit of prediction-time information, split integrity, cell-block uncertainty, schema controls, threshold decisions, and localization sensitivity. Twelve analyses use 13,831 eligible records from 13,876 CALCE, NASA, and Oxford observations. A pooled-domain, cell-held-out baseline reaches 94.8% coverage (cell-block lower bound 94.2%). Across six cross-dataset tasks, method-matched removal of the most recent measured SOH yields statistically supported coverage reductions in five tasks, by as much as 60.4 percentage points; the remaining task and pooled-domain baseline are compatible with no effect. The as-used schema bundle decreases leave-NASA coverage by 12.0 percentage points and increases false acceptance by 100.0 percentage points at the illustrative SOH cutoff 0.90 (exact paired-cell p = 9.54e-07); an independent control restricts interpretation to a bundle-level association. A localized-residual comparator recovers one point-coverage failure; its design carries no shift-robust guarantee. Each gate couples evidence requirements to claim bounds, yielding an executable pre-submission check demonstrated on three public battery benchmarks. Prospective applications are needed to establish detection performance.

Keywords: conformal prediction, uncertainty quantification, battery state-of-health, prognostics and health management, predictive maintenance, benchmark integrity, distribution shift

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1. Introduction

Data-driven prognostics and health management (PHM) increasingly supplies the evidence on which maintenance and serviceability decisions rest, and closing the gap between research practice and trustworthy deployment is a recognized priority for the reliability community [1, 2]. Lithium-ion battery state-of-health (SOH) estimation is a flagship case: high-impact data-driven estimators are routinely evaluated on public cycle-level datasets [3, 4], and uncertainty quantification is now treated as a first-class evaluation criterion for such models [5, 6]. Conformal prediction — with its finite-sample coverage guarantee — is entering reliability-engineering and battery pipelines [7, 8, 9]. Because SOH intervals feed threshold decisions about maintenance, replacement, and second-life use, interval miscalibration has direct decision consequences [10, 11, 12].

SOH uncertainty estimates are often evaluated through cross-dataset benchmarks, where a model calibrated on one collection is tested on another — a realistic proxy for the domain shift that deployed prognostics faces [13]. In maintenance- and reliability-facing uses, it is tempting to read those intervals as evidence that threshold decisions remain trustworthy under protocol shift. Yet dataset composition and feature choices, rather than model quality, can drive reported battery-prognostics performance [14, 15], and the reliability of machine-learning evidence is itself an engineering quantity that requires pre-deployment assessment [16]. We therefore treat benchmark construction and prediction-time information as part of the reliability claim itself.

We turn this premise into an explicit, governed audit protocol and demonstrate it on real cycle-level SOH records from the Center for Advanced Life Cycle Engineering (CALCE), the National Aeronautics and Space Administration (NASA), and Oxford. Within these three collections, we ask how much of the apparent validity of conformal SOH evidence depends on access to the most recent measured SOH, and what remains interpretable after schema, protocol, and decision-level checks.

Two bodies of work bear directly on this question, but neither audits the benchmark itself. Conformal methods in reliability engineering and battery state estimation concentrate on producing or adapting predictive intervals [7, 17, 8, 18, 9], while transfer and domain-generalization methods concentrate on making estimators survive distribution shift [13, 19, 20]. Non-exchangeability theory bounds how far conformal coverage can drift [21, 22], but it does not identify which measurement-validity channel a particular benchmark contains. Section 2 details this gap; our protocol addresses it.

The main contributions of this paper are as follows:

1. We define a benchmark-integrity audit protocol whose six governed gates cover the prediction-time information set, split integrity, cell-block un-

certainty, schema negative controls, threshold-decision consequences, and localization sensitivity.

2. We quantify recent-SOH access as a load-bearing, task-dependent measurement condition, separating a method-selection ablation from method-matched, row-matched contrasts.
3. We audit as-used schema-bundle sensitivity with multiplicity-adjusted intervals and an independently permuted-feature negative control, showing how the control gate confines observed bundle effects to a sensitivity warning rather than a field-specific mechanism attribution.
4. We report a corrected leave-NASA (LNAS) schema-bundle false-acceptance increase of +100.0 pp (percentage points) at threshold 0.90 (exact paired-cell p = 9.54e-07; 21 effective cells and 1,127 records), linking interval behavior to threshold-decision risk under the bundled, noncausal interpretation.
5. We bound residual sensitivity with a localized-residual comparator envelope, distinguishing empirical recovery from a formal covariate-shift guarantee.

Operationally, raw cross-dataset conformal coverage becomes battery SOH reliability evidence only after the benchmark's prediction-time information, split construction, uncertainty units, controls, decision consequences, and localization have been audited. The protocol makes that audit concrete and executable.

The remainder of the paper is organized as follows. Section 2 reviews related work on conformal prediction, battery SOH uncertainty, and benchmark integrity. Section 3 defines the audit design, its provenance, and its claim boundaries. Section 4 reports evidence on recent-label dependence, schema-bundle sensitivity, residual drift, decision consequences, and localized calibration. Section 5 discusses implications and limitations, and Section 6 gives concluding remarks.

2. Related Work

2.1. Conformal prediction: guarantees and their boundary conditions

Conformal prediction provides finite-sample predictive-coverage guarantees under exchangeability [23, 24, 25]. Conformalized quantile regression calibrates quantile-regression estimates to retain finite-sample coverage [26], and weighted conformal variants extend the guarantee to covariate shift when the shift mechanism can be estimated [27]. The boundary conditions of these guarantees are now well mapped. When calibration and test data are dependent or non-exchangeable, coverage deviates by an amount that theory can bound but not eliminate [21, 22], and ensemble batch prediction intervals extend conformal thinking to dependent time series [28]. Online and adaptive variants track arbitrary distribution shifts, but they require sequential feedback after each prediction [29, 30], which a one-shot cross-dataset evaluation does not provide.

A parallel line sharpens the distinction between marginal and conditional validity: exact conditional coverage is unattainable in general, and localized or reweighted calibration offers only partial remedies [31, 32, 33]. Conformal risk control generalizes coverage to other monotone risks, including decision-relevant error rates, with non-exchangeable extensions [34, 35].

These guarantees do not by themselves certify a reliability benchmark. If the information set includes recent labels, dataset identity, or bookkeeping fields unavailable in the intended decision setting, nominal coverage can describe the constructed benchmark rather than the intended threshold-decision problem. The localized-residual analysis used in this paper is a diagnostic inspired by this literature, not an implementation of density-ratio weighted conformal prediction.

2.2. Uncertainty-aware battery SOH and remaining useful life prognostics

Battery SOH and remaining useful life (RUL) studies have long reported probabilistic forecasts and prognostic performance under uncertainty [36, 37, 38], and real-application SOH estimation has a mature methodological literature [39]. Within the reliability-engineering literature specifically, data-driven battery prognostics is a flagship application: high-impact estimators define the benchmark conventions on CALCE-, NASA-, and Oxford-style cycle datasets [3, 4, 40]. An explicitly uncertainty-aware thread now reports quantile intervals validated on the same public datasets [41], Bayesian-calibrated SOH uncertainty under deployment shift [42], probabilistic degradation-trajectory health assessment [43, 44], and recent dynamical-model variants [45]. Label scarcity at deployment is a recognized constraint in this literature: adaptive prediction with few or no in-service capacity labels is studied as its own problem [46, 47], which makes the availability of a recent measured SOH a substantive modeling assumption rather than a technicality. Dedicated benchmarks and reviews now treat uncertainty quantification as a first-class evaluation criterion for deep-learning prognostics [5, 2, 6, 48].

Conformal prediction itself has entered this application class in three roles. It produces calibrated RUL and degradation intervals [7, 49, 8, 18, 9], adapts interval inference to corrupted inputs or engineering-design uncertainty [17, 50], and supports anomaly detection or width-aware calibration [51, 52]. Related industrial time-series work also documents the exchangeability violations that motivate such adaptations [53]. These methods improve or diagnose interval estimators while largely taking benchmark construction as fixed; our audit puts that construction to the test.

2.3. Cross-dataset evaluation, transfer, and domain shift in prognostics and health management

Transfer research pursues several remedies for distribution shift. Domain-adaptation methods move SOH or RUL estimators across batteries, operating profiles, and laboratory or field domains [54, 55, 56, 13, 19, 57]. Condition-aware and domain-generalization methods target fleet or protocol variation [58, 20], while adjacent reliability work mitigates shift in data-driven risk assessment [59]. These studies focus on remedies and generally treat the benchmark schema as a fixed prerequisite for comparison. We take the complementary audit-side view: before asking whether a model transfers, we ask whether the benchmark's information set, splits, feature schema, and decision endpoints support the reliability conclusion being drawn.

2.4. Benchmark integrity, leakage, and evaluation validity

Machine-learning reliability research warns that apparent predictive performance can be driven by unintended signals, undocumented data provenance, or data-splitting and feature-availability choices [60, 61, 62]. The battery and PHM literatures now contain concrete instances: feature selection can produce misleading battery-lifetime predictions [15], training-set size and composition — rather than model class — can drive reported capacity-estimation accuracy [14]. Explainability analysis reveals which features an SOH model actually exploits [63], and open benchmark suites codify the feature schemas whose integrity this paper audits [64]. Within reliability engineering, an evaluation-validity genre is emerging: simple baselines can expose what domain-shift benchmarks actually measure in condition monitoring [65], and pre-deployment reliability assessment of machine-learning evidence is treated as its own engineering problem [16].

We found no integrated audit protocol in the literature reviewed above that treats a conformal SOH benchmark itself as the object of reliability analysis. We therefore analyze the benchmark evidence directly: how conformal interval behavior and threshold decisions change across recent-label, split, schema-bundle, residual-drift, and SOH-threshold diagnostics. Six gates organize these diagnostics, each with an explicit evidentiary requirement and a corresponding bound on the admissible claim.

3. Methodology

We develop a governed benchmark-integrity audit protocol for conformal SOH evidence. Table 1 defines six gates, each linking a measurement condition to the evidence it requires and the consequence of failure. The protocol grew from a sequence of completed diagnostic analyses. Component decision rules were recorded in the corresponding experiment plans before execution; the six-gate synthesis was developed afterward and is therefore not prospectively registered. A failed gate narrows the resulting claim without discarding a negative or heterogeneous result. The remainder of this section defines the protocol's gates, decision rules, and interpretation boundaries.

We organize the evidence around governed diagnostic analyses and generate every reported value from a scripted results table. All active rows are real battery state-of-health records from CALCE, NASA, and Oxford [66, 67, 68]; the analysis contains no synthetic records. The methodological contribution is the

Table 1: Governed benchmark-integrity audit protocol for conformal SOH evidence. Each gate names the measurement condition examined, the evidence required, and the consequence of failure applied in this audit.

Gate	Evidence required	Consequence of failure
G1 Information set	State whether recent measured SOH or another oracle label is available at prediction time	No claim for settings without recent labels
G2 Split integrity	Verify task-appropriate disjoint training, validation, calibration, and test indices; require cell-disjoint histories where the task claims cell-level holdout	Treat coverage as split-contingent
G3 Uncertainty units	Report cell-block uncertainty rather than row-level i.i.d. intervals	Treat uncertainty as optimistic
G4 Schema sensitivity	Compare schema contrasts with an independent permuted-feature control and multiplicity adjustment	Bundle-level warning; no mechanism claim
G5 Decision consequences	Report threshold false acceptance and false rejection with record denominators, effective cell counts, and sparse-cell flags	Do not promote coverage to serviceability evidence
G6 Localization sensitivity	Compare standard and localized calibration as a sensitivity envelope	Failure: no empirical recovery; apparent success: point-coverage sensitivity only. Neither outcome authorizes a shift-robust claim without a guarantee-bearing method and verified assumptions

protocol used to audit the benchmark evidence constructed from these public cycle-level datasets. The 12 completed diagnostic analyses comprise a pooled-domain, cell-held-out coverage baseline, a recent-label ablation, a cross-protocol transfer diagnostic, a residual-bias screen, which tests whether calibration residual signs vary systematically with cycle position, a host-timing screen, which tests whether results shift with the wall-clock execution host, and a negative-control check. The remaining analyses comprise a method-family comparison, a threshold decision-utility audit, a schema-cleaning audit, a hard-regime audit, a schema-to-decision audit, and a localized-residual comparator. Supplementary Tables S1–S2 map every analysis to its gate, task, feature schema, method family, parameter grid, selection rule, seed, plan identifier, governed rerun, and deviation record. The archived plans and regeneration sources are preserved in the public replication repository.

The physical case spans three laboratory platforms rather than one nominally homogeneous fleet. The CALCE subset contains six CS2 (1.10 Ah) and six CX2 (1.35 Ah) small prismatic cells with $LiCoO_2$ cathodes. The archive documents constant-current/constant-voltage charging for CS2 and 0.5 C or 1 C constant-current discharge for the selected CS2/CX2 cells; the retained row files do not encode ambient temperature [66]. The NASA subset contains 25 Li-ion cells from multiple Prognostics Center of Excellence batches; the repository describes charge, discharge, and impedance profiles for nominal 2-Ah cells,

Table 2: Real battery datasets after preprocessing. Rows are source/eligible; every eligible row has an earlier same-cell SOH record.

Dataset	Cell platform	Cycling / temperature	Cells	Rows	SOH range
CALCE	CS2/CX2 prismatic, $LiCoO_2$, 1.10/1.35 Ah	CS2: CC–CV charge; selected CS2/CX2: 0.5C/1C discharge; temperature absent	12	11,604/11,592	0.40–1.01
NASA	Multi-batch Li-ion cells; archive nominal 2 Ah; cathode not consistently encoded	Charge/discharge/EIS; retained metadata span 4–43 °C	25	1,753/1,728	0.41–1.00
Oxford	Kokam SLPB533459H4 pouch, 0.74 Ah	CC–CV charge; ARTEMIS urban-drive discharge; chamber 40 °C	8	519/511	0.62–1.00
Total	Three public laboratory platforms	Heterogeneous protocols and measurement conventions	45	13,876/13,831	0.40–1.01

and the retained metadata cover chamber temperatures of 4, 22, 24, and 43 °C, but do not consistently encode cathode chemistry [67]. Oxford contributes eight 740-mAh Kokam pouch cells aged in a 40 °C chamber with constant-current/constant-voltage charging and an ARTEMIS urban-drive discharge profile [68]. These differences make label and protocol comparability part of the audit boundary rather than a solved preprocessing detail. Table 2 summarizes the 45-cell source evidence base of 13,876 cycle rows and the 13,831 rows eligible for a strictly prior recent-SOH feature.

The seven evaluation tasks use the same model families but change the target-domain role. Task abbreviations use Main for the pooled-domain, cell-held-out task, and leave-CALCE (LCAL), target-recalibrated CALCE (TCAL), leave-NASA (LNAS), target-recalibrated NASA (TNAS), leave-Oxford (LOxf), and target-recalibrated Oxford (TOxf) for the six cross-dataset tasks. Leave-domain-out tasks fit, select, and calibrate on the other two domains and test on all eligible rows of the named held-out domain. Target-recalibrated tasks add early target-domain rows: progress 0–20% enters training, 20–35% calibration, 35–55% validation, and rows after 55% form the test set. These boundaries are pairwise disjoint. Table 3 reports the resulting row counts; the held-out target contains 12 CALCE, 25 NASA, or 8 Oxford cells, while Main uses 10 test cells. Main is a pooled-domain within-benchmark baseline: all three datasets contribute to fitting and test sets, while cell identities remain disjoint; it is not a single-physical-protocol validation.

The Main row counts follow governed cell partitions and the progress windows specified below. Leave-domain-out tasks reuse those partitions after restricting training, calibration, and validation to non-target domains; consequently, their source-domain rows need not account for every eligible source-domain row in Table 2. Target-recalibrated boundaries are time-forward and pairwise row-disjoint.

Main is a seed-42, domain-stratified cell split, not a within-cell forecasting split. Within each domain, shuffled cells are assigned 55%/20%/10%/remainder

Table 3: Operational task construction and row counts for training, calibration, validation, and testing.

Task	Construction	Test cells	Train/Cal/Val/Test rows
Main	Cell-disjoint mixed-domain held-out split	10	7961/880/680/2323
LCAL	Other domains to all CALCE rows	12	1122/241/116/11592
TCAL	Other domains + early CALCE to late CALCE	12	4566/1754/2344/5167
LNAS	Other domains to all NASA rows	25	7082/664/636/1728
TNAS	Other domains + early NASA to late NASA	25	12447/279/322/783
LOxf	Other domains to all Oxford rows	8	7718/855/608/511
TOxf	Other domains + early Oxford to late Oxford	8	13421/71/106/233

to training/calibration/validation/test; calibration retains progress at most 0.55 and test retains progress at least 0.35. The four cell sets are disjoint, and no test cell contributes an earlier row to model fitting, selection, or calibration. The complete cell membership is recorded below; the governed split manifest preserves row indices together with the manifest checksum.

Complete Main split membership (domain entries are row counts).

Train (7,961 rows; CALCE 6,839; NASA 879; Oxford 243): CALCE_CS2_33, CALCE_CS2_36, CALCE_CX2_33, CALCE_CX2_34, CALCE_CX2_36, CALCE_CX2_38, NASA_B0007, NASA_B0025, NASA_B0026, NASA_B0028, NASA_B0029, NASA_B0031, NASA_B0042, NASA_B0043, NASA_B0048, NASA_B0053, NASA_B0054, NASA_B0055, NASA_B0056, Oxford_Cell1, Oxford_Cell3, Oxford_Cell4, Oxford_Cell6.

Cal (880 rows; CALCE 639; NASA 216; Oxford 25): CALCE_CS2_35, CALCE_CS2_38, NASA_B0006, NASA_B0027, NASA_B0045, NASA_B0046, NASA_B0047, Oxford_Cell5.

Val (680 rows; CALCE 564; NASA 44; Oxford 72): CALCE_CS2_37, NASA_B0032, NASA_B0051, Oxford_Cell2.

Test (2,323 rows; CALCE 1,968; NASA 255; Oxford 100): CALCE_CS2_34, CALCE_CX2_35, CALCE_CX2_37, NASA_B0005, NASA_B0018, NASA_B0030, NASA_B0044, NASA_B0049, Oxford_Cell7, Oxford_Cell8.

For cell i and recorded cycle t, `prev_soh` is $SOH_{i,t-1}$: the immediately preceding recorded same-cell capacity-SOH after sorting by dataset, cell identifier, and cycle index. The $t-1$ measurement must occur before the prediction event at t. Each cell's first row has no such measurement and is excluded before every split is constructed (45 rows in total); no same-row target fill or other imputation is used. Thus, the recent measured SOH is an oracle only in the sense that it assumes a previous capacity measurement is actually available at prediction time, not because it accesses $SOH_{i,t}$.

The primary nominal coverage level is 90%. This paper reports coverage, width, dependence-aware uncertainty, threshold decision utilities, schema diagnostics, and localized-residual comparator outcomes from the scripted results table. Before the schema-to-decision analysis was executed, its experiment plan defined five contrasts across seven tasks and seven endpoints: one coverage endpoint plus the predeclared SOH thresholds 0.65, 0.70, 0.75, 0.80, 0.85, 0.90, giving 245 tests. Coverage and false-acceptance deltas in Tables 5 and 6 use Bonferroni-adjusted, SE-based normal intervals from 1000 cell-block resamples; the family-wise level 0.05 gives a per-test level near 2×10^{-4} and $z \approx 3.7$.

Zero-SE boundaries use the exact paired-cell sign test. By contrast, the Main 94.2%–97.1% interval and the lower bounds in Table 8 are 2.5/97.5-percentile cell-block bootstrap intervals, while Table 7 reports uncorrected 95% cell-block normal intervals. The limited number of battery-cell blocks makes extreme-tail normal widths uncertain; this is an additional reason to treat coverage flags as sensitivity signals and require a negative-control gate. The same data interface generates text, tables, and captions to avoid manual transcription.

The nominal target is the evaluation criterion, while each conformal variant selects its calibration quantile strictly on a disjoint validation split and never tunes on the test set. For the pooled-domain, cell-held-out diagnostic baseline, the selected operating point uses q-level 0.93 and realizes test coverage of 94.8%. We use this over-coverage as the baseline's 'apparent reliability' before domain shift or artifact removal and as the reference for subsequent measurement-validity checks against the 90% target.

We use persistence-anchor behavior as a diagnostic probe of recent-label and anchor dependence. Cross-protocol failures then become validity evidence: a failure that persists after schema cleaning is consistent with residual protocol drift, whereas a small, directionally heterogeneous, or control-inconsistent schema effect limits the schema inference. Recent-label dependence and residual drift cannot be fully separated in this evidence base, so residual drift remains an explicit boundary of interpretation.

The audit examines four diagnostic dimensions:

1. recent-label / anchor dependence;
2. as-used schema-bundle sensitivity;
3. residual protocol drift after schema checks;
4. SOH serviceability-threshold decision risk under finite-cell evidence.

The Main pooled-domain, cell-held-out diagnostic baseline is the selected persistence-anchor Mondrian conformal predictor [23]; the split-conformal constructions and coverage definitions follow standard practice [25]. The diagnostic families include quantile regression (QR), conformalized quantile regression (CQR) [26], and natural-gradient boosting (NGBoost) [69]. For the no-recent-SOH condition, validation selects a gradient-boosting domain-Mondrian conformal variant, a group-conditional calibration construction suited to settings where exchangeability across domains is not credible [21]. We use this condition to measure the benchmark's dependence on recent-label access.

The calibration-selection rule chooses the operating point within each condition, so the full-feature and no-recent-SOH conditions can select different conformal families. Their pooled-domain coverage difference measures the complete selection pipeline's response to removing recent labels, whereas the row-matched comparison in Section 4 supplies the same-method feature-isolation contrast. We use gradient-boosting CQR for decision-utility diagnostics, schema comparisons, and the localized-residual comparator. Training, validation, calibration, and test indices are pairwise disjoint within each task.

The as-used bundle contains `source_row_id`, `target_cycle_index`, and the permuted control column, whereas the clean-original schema removes all three. The bundle contrast therefore identifies their joint effect rather than a bookkeeping-specific mechanism. The independent control adds a fixed-seed (seed 1033) permutation of `log_cycle_index` to the clean schema and refits the complete pipeline on the same rows. Because this control uses one permutation draw ($B = 1$) rather than a sampled permutation null distribution, it serves only as a veto on mechanism attribution.

A decision-risk cell is defined relative to a capacity-SOH serviceability threshold and a predicted SOH interval with lower bound L and upper bound U for a test record with true SOH y. The record is threshold-acceptable when its lower bound L is at or above the threshold, threshold-unacceptable when its upper bound U is below the threshold, and uncertain when the interval straddles the threshold. A false-acceptance event is a threshold-acceptable record whose true SOH y is below the threshold; a false-rejection event is a threshold-unacceptable record whose true SOH y is at or above the threshold. The reported false-acceptance rate divides false-acceptance events by the number of actually threshold-unacceptable records, the false-rejection rate divides false-rejection events by the number of actually threshold-acceptable records, and the uncertain rate divides uncertain records by all records. Each rate carries a cell-block bootstrap confidence interval that resamples battery cells with replacement. When every contributing cell block contains zero events, that bootstrap has zero empirical variance; the table therefore marks the interval as bootstrap-degenerate rather than displaying a zero-width confidence interval. Such a boundary is not evidence of zero underlying risk. Auditing these monotone decision risks alongside coverage follows the same motivation as conformal risk control, which generalizes coverage guarantees to decision-relevant error rates [34]. These thresholds are capacity-based decision cutoffs, not demonstrated electrochemical or thermal safety boundaries. To avoid over-reading sparse cells, a small-denominator suppression rule flags any decision cell whose denominator is below 5 as exploratory rather than as a headline rate.

Schema-bundle sensitivity is adjudicated by a rule documented before the schema-to-decision analysis and applied identically to every task. A task is flagged when the as-used-minus-clean coverage interval excludes zero and the absolute delta is at least 3 pp (percentage points). This flag is necessary but not sufficient for a mechanistic claim: the independent permuted-feature control must remain null across both its coverage contrast and all six false-acceptance endpoints. The bundle's false-acceptance-rate delta is evaluated separately with multiplicity adjustment or, for zero-standard-error boundaries, an exact paired-cell sign test; cells below 5 effective observations are exploratory. This hierarchy prevents a bundled association from being relabeled as a bookkeeping mechanism, while still allowing a corrected bundle-level decision association to be reported without field-specific attribution.

Two dependence boundaries govern interpretation. First, the conformal calibration objects are cycle-level residuals from longitudinal cell records, so exchangeability depends on the task construction and does not extend automatically to independent operating histories. Under such dependence, theory bounds but does not eliminate deviations from nominal split-conformal coverage [22, 28]. We therefore resample cells as blocks and interpret the resulting intervals as dependence-aware diagnostics; a fully cell-independent prospective forecasting design would provide stronger evidence. Second, the main dependence probe treats recent-SOH as an oracle recent label. It is a legitimate autoregressive measurement in a laboratory setting where previous capacity has just been measured, but it may be unavailable to an online battery-management system. We therefore treat recent-SOH access as a measurement condition that must be declared, rather than labeling it as leakage in every setting.

The SOH label is capacity-normalized within the harmonized project data interface, following the conventions of published SOH benchmark suites on these datasets [64, 41], and Table 2 reports the ranges after quality assurance. Because the source datasets differ in cycling protocol, temperature range, and capacity measurement practice, label comparability forms part of the benchmark-validity boundary. We do not vary denominators, smoothing rules, or anomaly-removal policies here; attributing every cross-dataset coverage failure to physical protocol drift or model behavior would require that additional sensitivity analysis to be run.

To test whether a standard conformal failure is sensitive to residual localization, we also evaluate a heuristic localized-residual CQR comparator in the spirit of localized conformal prediction [32]. Calibration and target features are standardized by calibration means and standard deviations. For target row x, calibration score r_i receives Gaussian weight $w_i(x) = \exp\{-d_i(x)^2/(2h^2)\} + 10^{-12}$, where d_i^2 is the mean squared standardized distance over the recorded common shift features. The local radius is the weighted conformal quantile. Bandwidth $h \in \{0.25, 0.50, 1, 2, 4, \infty\}$ is selected only on validation data by the predeclared coverage-first-then-width objective; test metrics are not used. The comparator is motivated by covariate-shift conformal work [27], but it neither estimates a density ratio nor carries a formal covariate-shift guarantee. Meeting the envelope means that point coverage reaches nominal; a lower-bound shortfall is recorded only when the standard percentile lower bound is below nominal and the localized lower bound is at or above it. We use the result as a sensitivity diagnostic, not as a method comparison.

4. Results

We apply the audit protocol to a cross-dataset conformal SOH benchmark and examine the reliability evidence it produces. The analysis tracks how interval coverage and SOH threshold decisions change with recent-label access, split

Table 4: Gate-by-gate audit ledger. Later tables report interval estimates. Pass* denotes complete reporting with bounded interpretation; pp denotes percentage points.

Gate	Case evidence	Verdict	Claim bound
G1	Prior SOH declared; removal -60.4 pp in TCAL; TOxf +4.3 pp with interval including zero	Conditional	Exclude settings without prior labels
G2	Main cell-disjoint, governed source splits for leave-domain tasks, forward disjoint recalibration rows	Pass	Split-contingent
G3	Cell-block intervals, effective cell counts 8–25 across tasks	Pass	No row-i.i.d. precision
G4	LCAL/LNAS bundle effects, LNAS -12.0 pp, control non-null in LCAL/TCAL	Fail	Bundle only, no mechanism
G5	Both threshold errors, uncertainty status, cell/record counts, no sparse displayed cell	Pass*	Illustrative, not deployment rates
G6	TNAS recovers one point failure, Main/LCAL decrease, no lower-bound shortfall recovered	Bounded diagnostic	One point failure recovered; no lower-bound recovery or formal guarantee

construction, feature-schema sensitivity, residual protocol drift, and localized residual weighting.

The evidence base is a set of diagnostic contrasts over the same real-data benchmark splits. Task abbreviations follow Section 3: Main denotes the pooled-domain, cell-held-out task. Table 4 reports the argument-bearing contrasts needed to interpret reliability evidence; ancillary timing and residual-stress screens are treated as robustness checks rather than headline findings.

The selected Main baseline is the persistence-anchor Mondrian conformal diagnostic. On the pooled-domain, cell-held-out task it reaches coverage 94.8% with CI 94.2%-97.1%, mean width 0.038, and sample size 2,323. The ablation package performs a separate validation-only selection: its full-feature reference selects the persistence-anchor standard conformal diagnostic, with coverage 92.6% and mean width 0.027. Removing the recent-SOH feature changes that within-package selection to the gradient-boosting domain-Mondrian conformal diagnostic, with coverage 80.8% and mean width 0.221; relative to the stated ablation reference, the coverage change is -11.8 pp and the width increase is 0.193. Accordingly, these deltas use the ablation reference rather than the separately selected Main baseline. Figure 1 summarizes the audit logic: raw coverage is interpreted only after its measurement conditions are examined.

The cross-protocol results vary across uncertainty families and interval widths. Same-split QR/CQR/NGBoost comparisons show that higher coverage can require a different uncertainty family or wider intervals; for example, in the main held-out task, CQR is coverage-aligned but has 2.72 times the persistence-anchor width. In the common-cycle, no-recent-label audit, coverage is 92.9% on Main, 29.5% on LNAS, and 3.1% on LOxf. This heterogeneity rules out a universal monotone benefit of recent labels. On Main, the clean-original no-recent schema attains coverage of 92.4%. Because the as-used no-recent schema is flagged as contaminated in two tasks, we base the conclusions on row-matched

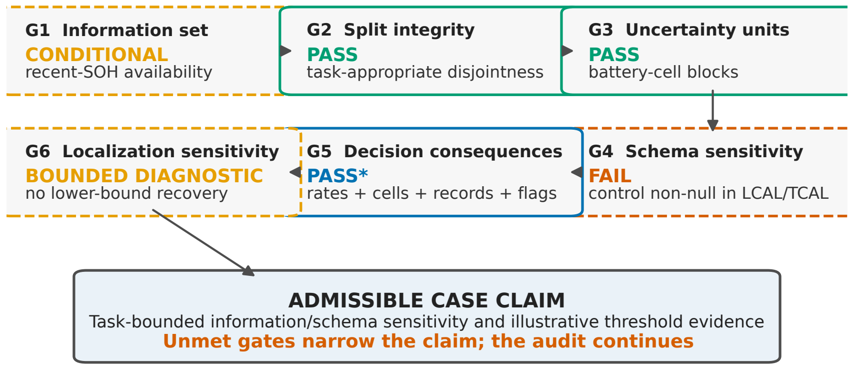

Figure 1: The six-gate benchmark-integrity audit converts an unmet evidence requirement into a narrower admissible claim rather than a stopped analysis. Case verdicts are shown inside the gates; dashed orange or vermilion borders mark conditional, bounded-diagnostic, or failed gates. Pass* means that G5's reporting requirements are met while its serviceability interpretation remains bounded.

Table 5: Task-wise coverage sensitivity (pp; Bonferroni-adjusted cell-block normal intervals). Cells/rows gives effective cells and test records. Recent is no-recent minus recent-SOH; Bundle is as-used minus clean; Control is the permuted-feature contrast. G4 fails for any non-null control endpoint.

Task	Cells/rows	No-recent − recent	Bundle	Control	Flag	G4 control
Main	10/2,323	+1.3 [-18.7, +21.4]	-4.1 [-21.6, +13.4]	-1.1 [-3.0, +0.7]	No	Pass
LCAL	12/11,592	-10.1 [-16.8, -3.3]	+3.2 [+1.2, +5.1]	-0.6 [-1.0, -0.2]	Yes	Fail
TCAL	12/5,167	-60.4 [-73.6, -47.2]	-2.2 [-6.9, +2.6]	-0.0 [-2.7, +2.6]	No	Fail
LNAS	25/1,728	-31.9 [-40.7, -23.1]	-12.0 [-20.5, -3.4]	-1.0 [-2.0, +0.1]	Yes	Pass
TNAS	25/783	-41.5 [-73.4, -9.7]	-10.0 [-31.2, +11.2]	-0.1 [-1.6, +1.3]	No	Pass
LOxf	8/511	-37.6 [-53.1, -22.0]	+0.4 [-0.5, +1.3]	-0.2 [-1.0, +0.6]	No	Pass
TOxf	8/233	+4.3 [-1.6, +10.2]	+0.0 [+0.0, +0.0]	-0.4 [-2.1, +1.2]	No	Pass

common-schema contrasts rather than the contaminated hard-regime rows. We retain the host-timing and residual-bias screens as robustness checks and draw no hardware or physical-degradation inference from them.

The method-matched recent-label delta is gradient-boosting CQR coverage under the common no-recent-SOH schema minus coverage with the recent-SOH reference, evaluated on identical task rows. This comparison is heterogeneous across tasks, which is central to the measurement-validity interpretation. Table 5 reports it beside the as-used schema-bundle contrast and its independent permuted-feature control.

The Control column in Table 5 displays the coverage endpoint. TCAL's G4 control failure instead occurs at the illustrative SOH cutoff 0.85, where the clean-original minus fixed-seed permuted-control false-acceptance contrast is -26.4 pp [-52.1, -0.8] after Bonferroni adjustment. The largest method-matched recent-label contrast occurs in TCAL, where removing the recent-SOH reference changes coverage by 60.4 pp; LNAS changes by 31.9 pp. Across the six cross-dataset tasks, five adjusted intervals support coverage reductions; the TOxf estimate is positive but its interval includes zero, as does Main's. The evidence therefore establishes task-dependent magnitudes, not a supported positive reversal. Because these comparisons hold the conformal family and task rows fixed, they isolate information-set sensitivity more cleanly than the selection-changing Main ablation described above. The Main ablation changes coverage by -11.8 pp, but its selection rule switches conformal families and therefore measures the response of the complete selection pipeline, not a same-method feature effect.

The as-used schema-bundle contrast passes the experiment-level coverage flag in 2 tasks: LCAL, LNAS. LNAS coverage changes by -12.0 pp; its maximum standardized mean difference falls from 7.15 in the raw schema to 1.32 in the common schema. The as-used bundle, however, jointly changes `source_row_id`, `target_cycle_index`, and the permuted control column, and the independent permuted-feature control is non-null in LCAL, TCAL. G4 therefore identifies bundle sensitivity but cannot assign it to a bookkeeping field: schema identity, finite-cell structure, and residual shift remain entangled. The failed control vetoes the stronger artifact-mechanism interpretation.

LNAS shows a multiplicity-adjusted schema-bundle false-acceptance increase of +100.0 pp at SOH threshold 0.90: the rate is 100.0% under the as-used schema and 0.0% under the clean schema over 1,127 actually threshold-unacceptable records and 21 effective cells (exact paired-cell p = 9.54e-07). The bundle's decision effect is directionally heterogeneous. The largest LCAL contrast is -100.0 pp at SOH 0.65 (0.0% as-used versus 100.0% clean; 1,936 records, 12 effective cells and 12 discordant cells; exact paired-cell p = 4.88e-04), which is risk-decreasing and does not survive multiplicity correction. Only LNAS passes correction, and the opposite LCAL direction shows that the bundle is not uniformly risk-increasing. Table 6 reports the largest observed eligible contrast for each task, and the complete 245-contrast ledger remains in the machine-readable results archive. We therefore report bundle-level decision associations rather than bookkeeping-specific effects or deployment-safety rates. Table 7 supplies the corresponding absolute decision risks, and Figure 2 visualizes the task-dependent information-set and schema-bundle sensitivities.

We report decision-utility rates with their denominators and cell-block intervals. False-acceptance and false-rejection events follow the Section 3 definitions. Table 7 reports representative decision cells at SOH threshold 0.80. It shows why coverage alone is insufficient: the leave-NASA CQR cell has a lower false-acceptance point estimate but leaves a large uncertain region, whereas NGBoost exhibits a high false-acceptance rate. These cells provide boundary evidence for the audit; operational use requires threshold-specific validation.

All representative G5 cells in Table 7 report both error directions and exceed the denominator-5 suppression floor. The two zero-event LNAS false-rejection cells are marked bootstrap-degenerate because cell resampling cannot estimate

Table 6: Largest eligible schema-bundle false-acceptance contrast by task (pp). As-used/clean gives the two base rates (percent); cells/rows gives the effective cell blocks and actually threshold-unacceptable records; discordant is the number of nonzero paired-cell differences. Intervals are Bonferroni-adjusted and are omitted where the cell-level standard error is zero; those cells instead show exact paired-cell p-values. Only LNAS passes correction.

Task	Delta [CI]	As-used/clean	SOH	Cells/rows	Discordant	Exact p	Corrected?
Main	+4.4 [-35.9, +44.6]	6.0/1.7	0.75	7/712	4	–	No
LCAL	-100.0	0.0/100.0	0.65	12/1,936	12	4.88e-04	No
TCAL	+21.5 [-25.9, +68.8]	28.6/7.1	0.85	12/5,036	9	–	No
LNAS	+100.0	100.0/0.0	0.90	21/1,127	21	9.54e-07	Yes
TNAS	+45.8 [-19.8, +100.0]	97.6/51.7	0.75	14/288	8	–	No
LOxf	+0.0	0.0/0.0	0.90	8/342	0	1.000	No
TOxf	+0.0	0.0/0.0	0.90	8/233	0	1.000	No

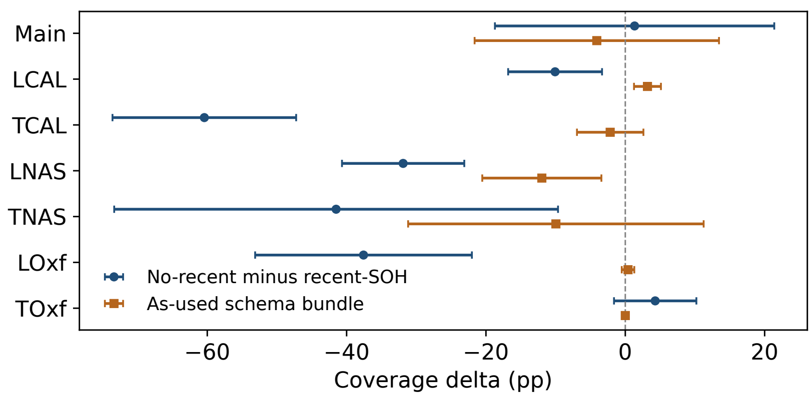

Figure 2: Per-task recent-label and as-used schema-bundle coverage deltas (percentage points) with Bonferroni-corrected normal intervals. Recent-label deltas are large and negative in several tasks; bundle deltas are smaller, with LCAL and LNAS meeting the coverage flag. The endpoint-wide negative-control gate fails in LCAL and TCAL. The dashed line marks zero.

Table 7: Illustrative threshold-0.80 decision cells (percent; uncorrected 95% cell-block CI). c/n gives effective cells/records; BD marks a bootstrap-degenerate all-zero boundary, and Sparse denotes any denominator below 5.

Task	Method	False acceptance	False rejection	Uncertain	Flag
Main	Persistence anchor	1.4 [0.1, 2.7]; c=9, n=1,270	2.1 [1.1, 3.1]; c=10, n=1,053	18.1 [12.0, 24.1]; c=10, n=2,323	OK
Main	CQR	1.1 [0.0, 2.4]; c=9, n=1,270	0.7 [0.0, 1.7]; c=10, n=1,053	30.9 [12.4, 49.4]; c=10, n=2,323	OK
Main	NGBoost	1.0 [0.0, 2.3]; c=9, n=1,270	6.9 [0.0, 19.2]; c=10, n=1,053	33.3 [25.3, 41.3]; c=10, n=2,323	OK
LNAS	Persistence anchor	1.9 [0.3, 3.5]; c=16, n=627	1.0 [0.1, 1.9]; c=25, n=1,101	6.1 [3.9, 8.4]; c=25, n=1,728	OK
LNAS	CQR	1.4 [0.1, 2.7]; c=16, n=627	0.0 [BD]; c=25, n=1,101	39.6 [28.2, 51.0]; c=25, n=1,728	OK
LNAS	NGBoost	69.2 [46.8, 91.6]; c=16, n=627	0.0 [BD]; c=25, n=1,101	11.5 [1.1, 21.8]; c=25, n=1,728	OK

Table 8: Heuristic localized-residual CQR coverage by task. Cells/rows gives the effective battery-cell blocks and test records. Std (low) and Local (low) give point coverage with the 95% cell-block lower bound in parentheses; Delta is local minus standard coverage (pp). Pt $\geq$ nom marks nominal point coverage, not a formal coverage guarantee.

Task	Cells/rows	Std (low)	Local (low)	Delta	Pt $\geq$ nom
Main	10/2,323	95.9% (95.0%)	89.4% (83.7%)	-6.5 pp	N
LCAL	12/11,592	75.5% (69.1%)	67.2% (60.9%)	-8.3 pp	N
TCAL	12/5,167	65.6% (52.9%)	68.0% (56.5%)	+2.4 pp	N
LNAS	25/1,728	90.4% (86.0%)	91.2% (87.0%)	+0.8 pp	Y
TNAS	25/783	88.0% (83.6%)	93.7% (89.7%)	+5.7 pp	Y
LOxf	8/511	42.9% (35.0%)	42.9% (35.0%)	+0.0 pp	N
TOxf	8/233	98.7% (96.9%)	98.7% (96.9%)	+0.0 pp	Y

unseen-event uncertainty from an all-zero boundary. G5 therefore passes as a reporting audit. Its application-level interpretation remains limited by the illustrative thresholds, the number of effective cells, and the inability of zero observed events to establish zero operational risk.

The localized comparator's coverage is below the nominal point target in Main, LCAL, TCAL, and LOxf. In LCAL, the standard comparator has coverage 75.5%, and localization changes coverage by -8.3 pp without repairing that failure. In Main, the standard comparator is above target at 95.9% with lower bound 95.0%; localization changes coverage by -6.5 pp and lowers the bound to 83.7%, thereby converting a pass into a point-coverage and lower-bound shortfall. TNAS is the only task in which localization converts a standard point-coverage failure into a pass; LNAS and TOxf already meet the point target under the standard comparator. No lower-bound shortfall crosses the nominal boundary from failure to pass. The comparator uses distance-based residual localization rather than density-ratio weighting, so its outcomes are empirical diagnostics without a covariate-shift coverage guarantee. We use Table 8 as a sensitivity envelope, not as a method ranking.

5. Discussion

The audit changes the interpretation of cross-dataset conformal SOH benchmarks. Coverage is conditional on the benchmark's measurement validity and information set; it is not automatically a reliability or serviceability result. Recent-label access can make a benchmark easier than the intended forecasting problem. The as-used schema-bundle contrast reveals an unstable evaluation design, but it jointly changes multiple columns, and the endpoint-wide independent-control gate fails in LCAL and TCAL. The corrected LNAS false-acceptance association is therefore retained at the bundle level without attribution to bookkeeping fields or promotion to a causal schema mechanism. Residual drift, finite-cell variation, and label-definition differences remain plausible contributors.

Our protocol addresses a different layer from current method development. Existing conformal work in prognostics and battery state estimation — split-conformal RUL pipelines, robust online conformal prediction, and conformalized quantile regression for electrified transportation — concentrates on producing better intervals [7, 17, 8, 18, 9], while transfer and domain-generalization research concentrates on making models survive shift [19, 20]. Neither line asks whether the benchmark used to certify those intervals measures the intended deployment question. Theory explains why coverage can degrade under non-exchangeability [21, 22], and industrial time-series studies document such violations [53]; the six gates locate the failure channel within a particular benchmark. Adaptive and online conformal methods require sequential feedback that a one-shot cross-dataset evaluation does not provide [29, 30]. The one-task recovery by the localized-residual comparator is likewise consistent with localized-conformal work: reweighted calibration can improve conditional behavior where local residual structure is informative, but it offers no general conditional guarantee [32, 33].

For reliability and battery-health practitioners, the results call for a pre-submission benchmark-integrity discipline. Prognostic intervals feed predictive-maintenance planning [10, 70], condition-based policies [11, 71], and multi-component scheduling [72, 73]; battery-health predictions also guide optimized maintenance actions [12]. The observed +100.0 pp false-acceptance change shows how an unaudited benchmark can misprice risk at a serviceability threshold. Estimating the corresponding fleet cost would require application-specific threshold validation. The opposite-direction -100.0 pp LCAL contrast also shows that the bundle is not uniformly risk-increasing. Analysts should declare recent-SOH availability at prediction time, demonstrate task-appropriate split integrity, use cell-block uncertainty, test schema claims against independent negative controls, and report threshold decisions with effective cell counts and record denominators. Localization is useful as a sensitivity test; reliability still depends on a guarantee-bearing method whose assumptions match the task. Table 1 condenses this discipline into six governed gates and extends the reliability-engineering practice of assessing machine-learning evidence before use [1, 16, 65, 48] to conformal SOH benchmarks.

Recent-label effects are not monotone. Method-matched deltas are strongly negative in TCAL, LNAS, TNAS, and LOxf, smaller but still negative in LCAL, near zero in Main, and positive in TOxf. This reversal matters because recent-SOH changes the estimand and interacts with the target task rather than acting as a universally helpful covariate. The practical failure is undeclared information availability. Label-scarce deployment studies make that failure concrete: adaptive battery prognostics cannot assume in-service capacity labels [46, 47], and feature-availability choices can mislead battery-lifetime prediction [15]. A benchmark that uses an oracle recent label needs a separate evaluation before it can support a no-recent-label deployment claim.

The localized-residual comparator sharpens this interpretation. It changes empirical coverage in several tasks and recovers exactly one standard point-coverage failure, while other tasks remain below target. The result shows sensitivity to local residual composition, but the absence of density-ratio weighting precludes a shift-robust coverage claim.

External breadth is the main limitation. The evidence comes from CALCE, NASA, and Oxford and therefore establishes sensitivity only within these benchmark families. We excluded the Massachusetts Institute of Technology-Stanford-Toyota fast-charge dataset [74] because its lithium-iron-phosphate chemistry and fast-charge-to-failure protocol would add chemistry and protocol confounding, and because its records did not pass the same preprocessing and schema-compatibility checks. External validation on broader modern laboratory and field datasets is needed.

Several design limitations follow directly from the audit framing. Calibration residuals are cycle-level observations from longitudinal cell histories; cell-block intervals quantify rate uncertainty but do not create independent exchangeable histories. A stronger follow-up should use cell-disjoint prospective splits, report row-weighted and cell-macro coverage, and show worst-cell behavior near serviceability thresholds. This paper treats the recent measured SOH as an oracle recent label; deployment-facing work should compare oracle, estimated, and unavailable recent-SOH on identical rows. The capacity-based label is harmonized only through the current preprocessing interface. G4's independent control uses a single permutation draw ($B = 1$), so it cannot characterize a permutation null distribution. Because the control acts only as a veto, a non-null draw can withhold mechanism attribution but cannot create it; repeated permutation draws would quantify how often a null feature produces a control effect as large as the LCAL result. Likewise, the zero-event false-rejection cells in Table 7 are bootstrap-degenerate boundaries rather than evidence of zero underlying risk. Moreover, the original frozen processed CSV could not be recovered byte-for-byte; the repaired analyses were regenerated from the preserved raw holdings under Python 3.11.15 and scikit-learn 1.7.2. Although raw split membership was reproduced, this provenance discontinuity limits exact numerical replication of the earlier exploratory run and is one reason the present conclusions rely only on the corrected rerun.

The audited SOH thresholds (0.65, 0.70, 0.75, 0.80, 0.85, 0.90) are illustrative capacity-SOH cutoffs used to demonstrate the diagnostic quantities. Maintenance, warranty, or end-of-life use would require application-specific validation; the cutoffs are neither electrochemical safety boundaries nor thermal-runaway risk thresholds. General battery-management validation will require field data, prospective forecasting splits, and application-specific risk models.

6. Conclusion

This paper develops and demonstrates a governed benchmark-integrity audit protocol for conformal battery SOH evidence. In the corrected real-data analyses, method-matched recent-label removal produces statistically supported coverage reductions in five of six cross-dataset tasks, with task-dependent magnitudes, while the remaining task and the pooled-domain baseline remain compatible with no effect. The protocol also detects that the as-used schema bundle changes coverage in two tasks and increases LNAS false acceptance at one threshold after multiplicity correction. The LCAL decision contrast has the opposite sign and does not pass multiplicity correction. Because the bundle jointly changes multiple columns and the endpoint-wide independent control is non-null, the control confines interpretation to a bundle-level association rather than a field-specific mechanism.

The resulting contribution is fivefold. First, the paper defines an explicit six-gate benchmark-integrity protocol. Second, it quantifies task-dependent recent-label sensitivity with method-matched contrasts. Third, it combines multiplicity-adjusted schema contrasts with an independent negative control. Fourth, it reports a corrected LNAS link between schema-bundle sensitivity and threshold false acceptance. Fifth, it uses a localized-residual sensitivity envelope to distinguish empirical recovery from a covariate-shift guarantee.

Each gate ties a measurement condition to an evidentiary requirement and a bound on the resulting claim. The gates are formulated independently of the three datasets, but this case establishes executability and claim control rather than independent detection performance. The next validation step is prospective application to benchmarks we did not construct.

Within the three public laboratory datasets studied here, the implication for reliability engineering is direct: coverage should support a serviceability interpretation only after the benchmark's information set, split integrity, uncertainty units, controls, decision consequences, and localization sensitivity have been audited. Broader field data, prospective forecasting splits, and repeated-control designs remain necessary before generalizing beyond this case.

Data Availability

This study uses real cycle-level SOH records from three public battery datasets: the CALCE Battery Research Group data repository [66], the NASA Ames Prognostics Data Repository [67], and the Oxford Battery Degradation Dataset [68], all accessed 2026-06-02. The versioned derived-data package, source-data provenance, split manifests, run manifests, archived analysis plans, and regeneration code underlying every reported value are publicly available at https://github.com/QuantDiviner/NEX-CP-conformal-soh-audit. Supplementary Tables S1–S2 provide the case-implementation and provenance matrix.

Code Availability

The complete analysis code, locked dependencies, and run manifests are publicly available at https://github.com/QuantDiviner/NEX-CP-conformal-soh-audit. The repository regenerates the scripted results table from which all manuscript values are drawn.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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During the preparation of this work, the authors used OpenAI Codex for code editing assistance, manuscript formatting, and language polishing. The authors subsequently reviewed and edited the content, verified all reported evidence and results, and take full responsibility for the content of the publication.

CRediT Authorship Contribution Statement

Qingsong Shan: Software, Data curation, Formal analysis, Validation, Writing – review & editing. **Qianning Liu**: Conceptualization, Methodology, Investigation, Visualization, Writing – original draft, Writing – review & editing, Funding acquisition.

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